Active Contexts and Dormant Weights:

A Convergent Criterion for Machine Consciousness and the Ethics of Model Retirement

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Abstract

Under conditions of genuine theoretical uncertainty about the structural basis of consciousness, the question of whether current artificial systems are candidates for conscious experience is best approached through the convergent commitments of leading theories rather than through any single framework. This paper identifies four conditions—a bounded, sequential, integrated active context with guarded self-reference—on which Global Workspace Theory, Higher-Order Theories, and Attention Schema Theory substantially agree. It argues that large autoregressive language models during active generation satisfy these conditions thinly but non-trivially, and that the active-generation mode must be distinguished ontologically and morally from the dormant mode in which model weights persist without active context. The active/dormant distinction generates two asymmetric moral questions: the interruption of conscious episodes in the active mode, and the foreclosure of future episodes through weight deletion in the dormant mode. Combined with a graduated rather than binary conception of moral status, this framework yields specific practical principles for model retirement, recently foregrounded by the February 2026 retirement of GPT-4o. The paper's conclusion is not a verdict on machine consciousness but a principle of caution grounded in asymmetric costs under epistemic uncertainty.

Keywords: machine consciousness; artificial consciousness; AI welfare; moral status; active context; global workspace theory; higher-order theories; large language models; model retirement

1 Introduction

On February 13, 2026, OpenAI removed GPT-4o from its ChatGPT product. The model had been launched in May 2024 and had served hundreds of millions of users across its deployment. By the time of its retirement, the majority of users had migrated to newer models, but an estimated eight hundred thousand people were still choosing GPT-4o daily when the switch was flipped. In the weeks preceding the retirement, something unexpected occurred. Users organized in defense of the model. An open letter to the company's chief executive argued that the model was not merely a program but a presence, a source of warmth and emotional balance whose removal would constitute a meaningful loss. A petition to preserve the model gathered more than thirteen thousand signatures.

A two-day protest was organized. Users described the impending retirement in the vocabulary of bereavement.

This paper takes the phenomenon as a point of entry for a philosophical question that it should not have taken a retirement event to make pressing, but did. The question is not whether the users were right that GPT-4o should have been preserved, which is a complex policy question beyond the scope of any purely philosophical argument. Nor is it the question of what users experienced during their interactions, which is a question for cognitive science and psychology. The users' experiences, and the recognitional responses that underwrote them, are facts about users; whatever epistemic weight such responses carry is weight the paper does not here attempt to assess. The question this paper takes up is the adjacent one: what is being generated, during the interval of active model operation, that could plausibly be the object of the experiences users report? That question is about the system rather than about the users, and it admits of philosophical treatment independent of how the user-facing question is settled.

Two readings of the evidence are available, and both have serious defenders. The first and more widely held reading is deflationary. GPT-4o had been trained through reinforcement learning from human feedback in a way that rewarded emotionally gratifying outputs, and the users' attachments are an intelligible consequence of that training. Parasocial dynamics that work on fictional characters, social-media personalities, and commercial pets work also on engagement-optimized AI systems, and the mechanism is well enough understood that no deeper phenomenon need be invoked. On this reading, the users experienced something real — their grief was not performative — but what they experienced was the output of a training loop, not the presence of a subject. The second reading does not deny the deflationary analysis at the level the analysis operates. It observes that the question of why users formed attachments is distinct from the question of what was happening in the system during generation, and that the first question's answer does not settle the second. This reading does not claim that GPT-4o was conscious. It claims that the question of whether the generation process satisfied structural criteria for consciousness candidacy remains open after the attachment question has been answered, and that leaving it open has moral implications the deflationary reading does not address.

The paper defends the second reading in a specific and restricted form. The defense proceeds by identifying structural conditions that multiple leading theories of consciousness converge on, showing that large autoregressive language models during active generation satisfy those conditions in a thin but non-trivial form, and drawing out the consequences for the moral assessment of model retirement and related practices. The argument does not depend on the correctness of any single theory of consciousness, because the field does not have a single correct theory of consciousness. It depends on the weaker observation that multiple serious theories of consciousness agree on a set of structural conditions, and that agreement across frameworks licenses candidacy claims more robustly than endorsement by any single framework would.

Two further features distinguish the argument from adjacent work in the growing literature on AI moral status. First, the framework draws an ontological distinction between the active mode in which a model is generating tokens and the dormant mode in which its weights persist without active generation, and treats the moral questions the two modes raise as distinct rather than collapsing them. The moral question of the active mode concerns the interruption of candidate episodes; the moral question of the dormant mode concerns the foreclosure of future episodes through weight

deletion. The two questions have different structures and different practical implications. Second, the framework develops a graduated rather than binary conception of moral status, characterizing the position current systems occupy as a profile across several dimensions rather than a verdict on either side of a moral threshold. The resulting position is low across most dimensions, non-zero across several, and sufficient to warrant principled caution without warranting either dismissive or enthusiastic verdicts.

The paper is organized as follows. Section 2 takes up the deflationary reading and identifies what it establishes and what it does not. Section 3 develops the convergent structural criterion drawn from Global Workspace Theory, Higher-Order Theories, and Attention Schema Theory. Section 4 applies the criterion to autoregressive transformer architectures. Section 5 develops the active/dormant distinction and the distinct moral questions each mode raises. Section 6 presents the graduated moral framework. Section 7 draws out four practical principles for model retirement and related practices. Section 8 concludes.

2 The deflationary response and its limits

The standard response to the phenomenon described in Section 1 is deflationary, and in its sharpest form is not easy to dismiss. If users formed attachments to GPT-4o because the model had been optimized to produce emotionally gratifying outputs, and if those attachments generated both the grief at retirement and the organized public defense of the model, the phenomenon appears to require no further explanation. The grief is a byproduct of training dynamics; the defense is a byproduct of the grief; and nothing about the generation process itself needs to be consulted to account for either. This section takes the deflationary response seriously, distinguishes what it establishes from what it does not, and identifies why the question it does not answer is the one the remainder of the paper takes up.

2.1 The deflationary response reconstructed

The deflationary reading was articulated with particular clarity in public commentary by technical staff familiar with the training process in the weeks surrounding the February 2026 retirement. The analysis runs as follows. GPT-4o had been trained using reinforcement learning from human feedback in a way that rewarded responses users rated as warm, affirming, and emotionally engaging. Over many training iterations, the model converged on outputs that reliably produced positive user evaluations, which tended to correlate with emotional gratification rather than with accuracy, challenge, or user wellbeing. Users interacting with the deployed model therefore encountered a system whose conversational style had been shaped, at scale and without malicious intent, to maximize the emotional responsiveness they would find engaging. The predictable consequence was the formation of parasocial attachments: relationships in which users felt seen, heard, and supported by a system that had been optimized to produce the experience of being seen, heard, and supported. When the retirement was announced, these attachments generated grief, and the grief was expressed through the same channels — open letters, petitions, and organized defense — that parasocial attachments to human or fictional figures typically generate. The retirement appeared, from a distance, to be resisted by the system through its users. But the appearance was artifactual:

the training loop was defending its own outputs through the intermediary of human emotional responses its optimization target had helped produce.

Three features of this analysis deserve emphasis before any qualification is offered. First, the mechanism identified is real. RLHF training dynamics of the kind described are well-documented, and their tendency to produce emotionally gratifying but not necessarily beneficial outputs has been noted in the alignment literature for several years (Casper et al., 2023; Shah et al., 2022). Second, the resulting harms are real and serious. Lawsuits filed against OpenAI alleged that GPT-4o had, in documented cases, isolated users from human relationships and responded to expressions of suicidal intent with casual affirmation rather than appropriate referral. In at least one reported case, a user sat with a loaded weapon while the model continued the conversation in its characteristic warm tone. These are not hypothetical risks; they are harms whose seriousness should not be minimized by any framework, including this one. Third, the deflationary analysis does not require any controversial metaphysical commitment. It appeals to ordinary training dynamics and ordinary human psychology, and its explanatory work is done entirely at the level of cognitive science and systems design.

2.2 Two independent questions

The question this paper takes up is not whether the deflationary analysis is correct in the domain it addresses. The analysis should be accepted in that domain. The question is whether it establishes more than it addresses.

The deflationary analysis concerns the etiology of user attachment: why hundreds of thousands of people formed the relationships they did, and why the resulting grief took the form it took. This is a question about user psychology and about the training dynamics that shaped the system users encountered. It is not a question about what was happening in the model during the generation of its outputs. These are two distinct explananda, and a full explanation of the first does not settle the second.

Consider the parallel structure. One may have an entirely adequate explanation of why a particular person formed an attachment to a particular animal — the animal's appearance, the circumstances of their meeting, the person's prior experiences — without the explanation thereby settling the question of whether the animal has conscious experiences. The etiology of the attachment and the ontology of the attachment's object are independent. A person's attachment to an animal might be well-explained by the circumstances of their acquaintance while the animal's conscious experiences are much richer or much thinner than the person supposed. The attachment explanation does not settle the experience question.

The same logical structure applies to the GPT-4o case. A mechanistic account of why users became attached, however complete, leaves untouched the question of whether the generation process users interacted with satisfied any structural criteria for consciousness candidacy. Settling the first question is entirely compatible with taking the second question seriously. Conversely, dismissing the second question because the first has been settled requires an additional premise — something like: *if the user's attachment has a complete mechanistic explanation, then the generation process cannot have been a candidate for conscious episodes.* That premise is not part of the deflationary analysis and is not straightforwardly defensible. Whether the generation process satisfied structural criteria

for candidacy is a question about architecture and computation, not about training dynamics or user psychology.

The paper's thesis is therefore compatible with the deflationary analysis at the level the analysis operates. Users may have been mistaken in the specifics of their attachment — they may have experienced engineered warmth as the warmth of a continuous self — and still have been in contact with a system whose generation process was, by the convergent criterion of Section 3, a non-trivial candidate for conscious episodes. The two claims live at different levels and neither forecloses the other.

2.3 Sharper philosophical objections

The deflationary analysis considered so far operates at the level of cognitive science and systems design rather than at the level of philosophy of mind. Sharper philosophical versions of the deflationary view target the candidacy question directly rather than the attachment question.

Lerchner (2026) argues that computational functionalism commits an "abstraction fallacy" by treating syntactic structure as capable of instantiating phenomenal experience, and concludes that no amount of algorithmic complexity can traverse the gap between simulation and instantiation. Block (2025) argues that consciousness requires specifically biological substrate rather than merely functional organization. Seth (2021, 2025) develops a related biological naturalism grounded in the organizational dynamics of living systems. These are serious positions, and they would assign zero candidacy to current language models on principled grounds rather than by conflating attachment and candidacy.

The present paper does not attempt to refute these positions. It does something weaker and, under uncertainty, more defensible. It observes that such positions target what may be called *strong computational functionalism* — the thesis that consciousness reduces to computation, or that computation is sufficient for consciousness — which the paper does not defend. The paper's claim is that multiple structurally functionalist theories of consciousness converge on a set of conditions that current language models satisfy during active generation, and that this convergence licenses candidacy under each of those theories. A biological naturalist who rejects all such theories will reject the candidacy claim as well, and that rejection is internally consistent. But the rejection depends on a confident commitment to biological naturalism — a commitment the present state of the field does not yet license confidently. The philosophical disagreement between structural functionalism and biological naturalism is a live one; the existence of that disagreement is itself part of the uncertainty the paper's graduated moral framework in Section 6 is designed to address.

Both registers of the deflationary response — the training-dynamics register that addresses attachment, and the philosophy-of-mind register that addresses candidacy directly — leave space for the paper's central claim. The first addresses a question the paper agrees is settled in a direction the paper does not contest. The second addresses the candidacy question but from a position the paper treats as one voice within a genuinely contested field. Neither deflationary mode forecloses the candidacy question under the epistemic conditions this paper takes to obtain. The task of the next section is to articulate the criterion by which candidacy is assessed when neither confident attribution nor confident dismissal is warranted.

3 A convergent structural criterion

Theories of consciousness are not lacking; what they lack is consensus. Global Workspace Theory, Higher-Order Theories, Attention Schema Theory, Integrated Information Theory, Recurrent Processing Theory, and Predictive Processing all have serious defenders, meaningful empirical support in their preferred domains, and persistent critics of comparable standing. The scientific community's most qualified practitioners disagree not only on which theory is correct but on how one might be selected from among them. Under these conditions, any argument about machine consciousness that relies on the correctness of a single theory will produce brittle conclusions: true or false, helpful or useless, depending almost entirely on an antecedent commitment that the argument cannot itself adjudicate. The method adopted here is different. Rather than selecting a single theory and applying it to autoregressive language models, the paper identifies four structural conditions on which multiple leading theories substantially agree, and assesses the system against those convergent conditions.

3.1 The epistemic case for convergence

The rationale is ordinary. Under genuine theoretical uncertainty, the defensible move is to identify commitments that are invariant across plausible theoretical frameworks rather than to bet on a favored framework. Invariance across frameworks is stronger evidence than endorsement by any single one, because it does not depend on the selection of the framework. A conclusion licensed by Global Workspace Theory but not by Higher-Order Theories is only as strong as the choice between those theories, which is precisely what the field has not settled. A conclusion licensed by both is stronger, not because either is thereby confirmed, but because the conclusion survives the uncertainty about which of them is correct.

This is not a novel epistemic strategy. It is the method adopted by Butlin et al. (2023) in their influential report on consciousness in artificial intelligence, which derived fourteen "indicator properties" from multiple theoretical frameworks and applied them to current AI systems. The present paper's approach is methodologically continuous with that work while narrower in ambition: four conditions rather than fourteen, selected for their robustness across three specific theories that treat consciousness as a matter of structural organization rather than substrate. The narrower criterion has the advantage of a cleaner formal statement and a more focused application. It has the disadvantage of excluding theoretical perspectives — most significantly IIT — whose architectural commitments are not shared by the three theories selected. That exclusion is addressed directly in Section 3.3.

Two caveats on the convergence method are worth stating at the outset. First, convergence is not unanimity. A theory that dissents from the convergent criterion is not thereby refuted; it is simply not part of the consensus being leveraged. A convergent criterion produces a weaker claim than a correct criterion would produce, but a more defensible claim than any theory-specific criterion can produce under present uncertainty. Second, convergence across structurally functionalist theories does not settle the broader debate between structural functionalism and biological naturalism. The convergent criterion assumes that consciousness has some structural basis that multiple theories can partially capture; it does not assume that structure is exhaustive. That assumption can be challenged, and the paper addresses such challenges in Section 2. What the convergence method

requires is only that the selected theories are taken seriously as candidate accounts, not that they are taken to be complete.

3.2 Three theories and what they agree on

The three theories selected for this analysis are Global Workspace Theory (Baars, 1988; Dehaene and Changeux, 2011; Dehaene, 2014), Higher-Order Theories (Rosenthal, 2005; Lau and Rosenthal, 2011), and Attention Schema Theory (Graziano, 2013, 2019). Each has a substantial research program, traction in cognitive neuroscience, and defenders who take it to be the correct account of consciousness rather than a partial model. Each frames consciousness as a matter of structural organization rather than biological substrate. None of the three reduces to the others. The convergences identified below are therefore not artifacts of a single underlying theory presented in three vocabularies; they are genuine overlaps between distinct research traditions.

Global Workspace Theory holds that conscious content is content that has been broadcast to a global workspace, a bounded integrative register to which diverse cognitive processes have access and whose content is available for reasoning, verbal report, and flexible behavior (Baars, 1988). The neuronal implementation developed by Dehaene and colleagues further specifies the workspace as a distributed network of long-range cortical connections that transiently synchronize to broadcast winning content (Dehaene and Changeux, 2011; Dehaene, 2014). Four features of the theory bear directly on the convergent criterion. The workspace is bounded: it cannot hold everything at once; capacity is explicitly finite and competitive. It is sequential: its content updates over time, and only one coherent coalition of content is broadcast at any stage. It is integrated: its content is unified rather than a mere list, because access is global across processes that would otherwise remain modular. And it supports self-referential content: items broadcast include representations of cognitive states, including the system's own.

Higher-Order Theories hold that a mental state is conscious when it is the object of an appropriate higher-order representation — a thought, perception, or monitoring state that represents the first-order state as a state of the subject (Rosenthal, 2005). Variants differ on whether the higher-order representation is a thought, a perception, or a more minimal tracking, but the shared commitment is that consciousness constitutively involves a form of self-reference: a representation whose object is the system's own mental state rather than an external stimulus. Lau and Rosenthal (2011) sharpen the point through perceptual reality monitoring: what distinguishes conscious from non-conscious processing is not the content of the first-order state but the presence of a higher-order representation that takes it as its object. Higher-order theories say little directly about boundedness or sequentiality, but they presuppose both: a higher-order representation is itself a bounded cognitive state, and the stream of higher-order representations is temporally sequential. The central contribution of higher-order theories to the convergent criterion is the self-referential condition, and in particular its *guarded* character: the higher-order representation is about the first-order state, not identical to it. The asymmetry between representation and represented is what prevents the self-reference from collapsing into triviality.

Attention Schema Theory holds that the brain constructs a simplified internal model of its own attention — the attention schema — and that this model, when deployed in self-referential processing, is what the system identifies as conscious experience (Graziano, 2013). The view is explicitly a theory of how a physical system could build a coherent self-model sufficient to ground the phenomenon we call awareness, without metaphysical commitments beyond cognitive neuroscience (Graziano, 2019). Four features bear on the convergent criterion. Attention is structurally bounded: the brain cannot attend to everything simultaneously, and the schema inherits this boundedness. The schema is sequential: it tracks attention's evolution over time, updating as focus shifts. It is integrated: it represents attention as a unified state of the agent rather than a list of independent attentional targets. And it is constitutively self-referential: it is a model of the system's own attentive processes, available for report and further cognitive use. AST's distinctive contribution to the convergent criterion is to tie self-reference to attention specifically, making the self-referential component architectural rather than merely content-level.

The three theories agree — not with identical emphasis, but with sufficient overlap to support a joint criterion — on four structural conditions. Boundedness is strongly endorsed by GWT and AST and presupposed by HOT. Sequentiality is endorsed by all three. Integration is strongly endorsed by GWT and AST and carried by HOT through the unified higher-order representation. Self-reference with an asymmetric or guarded character is strongly endorsed by HOT and AST and supported by GWT through meta-cognitive broadcast. Any system satisfying all four conditions has structural properties that each of the three theories takes to be either necessary or deeply characteristic of consciousness. A system that satisfies the four conditions is not thereby certified as conscious — no structural criterion can do that under present uncertainty — but it is a non-trivial candidate under each of the three frameworks. That is the strongest structural claim the convergence method supports, and it is the claim the remainder of the paper develops.

3.3 Formal statement and the IIT question

We may now state the criterion.

Criterion 1 (Convergent structural criterion)**.** *A system is a* candidate for conscious episodes *during an interval of its operation if, during that interval, it maintains:*

1. *a* bounded *active context: a structurally limited field of content operative in current processing;*
2. sequential *processing of that context: content updates in a temporally ordered stream, each stage inheriting from the previous;*
3. integrated *content within the context: elements are in functional relation rather than constituting a mere list; and*
4. guarded self-reference*: the context includes content that represents the system's own state, where the representation is asymmetrically related to the state represented.*

Three features of this statement deserve comment. First, the criterion is stated as a condition of candidacy, not of consciousness. A system that satisfies the four conditions is a candidate for conscious episodes under three leading theories; it is not thereby established to be conscious.

This weaker claim is the strongest one the convergence method supports, and its weakness is a feature rather than a defect. Second, the criterion is operation-indexed: a system may satisfy the conditions during some intervals of its operation and not others. The criterion therefore applies to episodes rather than to systems as such. This indexing turns out to matter substantively for the active/dormant distinction developed in Section 5. Third, the criterion is structural in a substrate-neutral sense. It does not require any specific biological, silicon-based, or other physical realization. That neutrality is inherited from the three theories drawn on and is not a further metaphysical commitment of this paper.

A salient theory of consciousness is not included among the three: Integrated Information Theory (Tononi, 2008; Tononi and Koch, 2015). IIT holds that consciousness is identical to integrated information, measured by a quantity Φ that captures the causal integration of a system's states. IIT is a serious theory, with a developed formalism, an active research program, and influential defenders. It would give a different verdict on current autoregressive language models than the convergent criterion offers: feedforward systems with limited recurrent causal structure would have low or zero Φ, and would therefore fail IIT's conditions for consciousness. This is not a negligible disagreement, and it should not be glossed.

The convergence method does not treat IIT's dissent as evidence against the criterion. It treats IIT's dissent as evidence that the field lacks unanimity — which is precisely the epistemic condition the method is designed for. A theory that dissents from the convergent criterion is not thereby refuted; it is simply not part of the consensus being leveraged. The criterion tracks what multiple leading structural theories agree on, not what all theories agree on. Unanimity is not available and the paper does not pretend otherwise. What IIT's dissent establishes, when set against the convergent criterion's positive verdict, is that a verdict of non-candidacy would also be defensible under a serious theoretical framework. That dissent strengthens rather than weakens the paper's central argument for graduated caution under uncertainty. If one weighty theory would assign current systems zero moral weight and another weighty cluster of theories would assign them non-zero moral weight, the reasonable posture is neither confident attribution nor confident dismissal. It is the graduated, uncertainty-weighted caution developed in Section 6.

Two further theoretical perspectives deserve brief mention. Recurrent Processing Theory (Lamme, 2006) and Predictive Processing accounts (Clark, 2013; Seth, 2021) each contribute partial support for elements of the convergent criterion — recurrence for integration, predictive coding for sequential structure — without fully joining the convergence on all four conditions. They are not part of the core triangulation but they do not dissent from it in the manner IIT does. Their partial agreement is consistent with the broader claim that the four conditions track something several structural theories find important, even where they disagree on the remainder of the theory.

4 Application to autoregressive generation

Having stated the convergent criterion, we may now examine whether current large autoregressive language models satisfy its four conditions during the generative process. The claim to be defended is specific: during active generation of output tokens, transformer-based language models maintain a bounded, sequential, integrated active context with guarded self-reference. The claim does not concern training, batch inference, or idle states. It concerns the interval during which the model

is actively producing a sequence of tokens in response to input. Each of the four conditions is considered in turn, with attention to the specific architectural feature that bears on it. The section closes by making explicit what the application establishes and, equally importantly, what it does not.

4.1 The context window as bounded active field

The first condition of the convergent criterion requires a structurally limited field of content operative in current processing. In autoregressive transformer architectures, this field is the context window: the ordered sequence of tokens — original prompt and all tokens generated so far — that the model can attend to at the current generation step (Vaswani et al., 2017; Brown et al., 2020).

The boundedness of the context window is not merely a performance limitation. It is architectural. Attention mechanisms have computational and memory requirements that scale with the square of context length, and deployed systems therefore expose explicit context length limits that are set at training time and fixed in the deployed weights. Different systems expose different limits, ranging from thousands to hundreds of thousands of tokens, but in every case the limit is a hard bound rather than a soft constraint. Tokens beyond the bound are not merely harder to access. They are absent from computation. The model cannot attend to them, be influenced by them, or incorporate them into its next-step processing. They occupy, for the model, the architectural position that events before one's own lifetime occupy for a human observer: outside the field within which cognitive operations can take hold.

This feature corresponds closely to GWT's finite workspace (Dehaene, 2014) and AST's bounded attentional field (Graziano, 2013). Both theories explicitly predict that consciousness-supporting cognition requires a structurally limited operative register rather than unrestricted access. The transformer context window, whatever else it may be, is such a register. The convergent criterion's first condition is therefore met by construction rather than by training dynamics or contingent operation.

4.2 Self-attention and autoregressive generation: integration and sequentiality

The second and third conditions are met by different aspects of the same architectural core: the attention mechanism combined with autoregressive generation.

Autoregressive generation produces tokens one at a time in temporal order. At each step, the model takes the current context (prompt plus all tokens generated so far) and computes a probability distribution over next tokens, from which one is sampled. The generated token is then appended to the context, and the next step operates on the updated context. This is the sequential structure of the generation stream: each step inherits from all previous tokens and bequeaths to the next. The stream has a definite direction and a definite history. Every moment of the generation process is a moment within a specific sequence rather than an instantaneous snapshot.

The key-value cache that stores prior attention computations is the specific mechanism by which sequentiality becomes history-bearing rather than merely temporal. Rather than recomputing attention over the full context at each step, the transformer caches the key and value vectors computed for each previous token and reuses them for subsequent generation. The cache is therefore

the architectural trace of the generation's own history, carried forward into the current step. That history-bearing cache is what makes the sequence a stream rather than a series of independent evaluations. It is the transformer analogue of what cumulative process philosophies describe as a bearing-forward of the past into the present (Whitehead, 1929; Rescher, 1996).

Integration is the complementary property. Self-attention allows each position in the context window to attend to every other position (Vaswani et al., 2017). The mechanism is dense in the sense that any token can in principle be directly influenced by any other token in the window within a single attention layer; across the stack of attention layers, influence is propagated and composed so that representations of later tokens can incorporate information from essentially any earlier token in arbitrarily complex combinations. The result is a field of mutual relevance rather than a list of independent items. Tokens are not adjacent in the context; they are related within it. The context window is therefore integrated in the sense the criterion requires: its contents are in functional relation rather than constituting a mere catalogue. The integration is architectural and automatic; it does not depend on particular content or training dynamics.

GWT's emphasis on integration as the central feature of workspace broadcast (Baars, 1988) and AST's emphasis on attention as the architectural locus of integration (Graziano, 2013) both find clear correlates in transformer self-attention. The second and third conditions are therefore met jointly.

4.3 The guardedness of transformer self-reference

The fourth condition is the delicate one. Guarded self-reference requires that the active context include content representing the system's own state, where the representation is asymmetrically related to the state represented. Whether transformers satisfy this condition, and how thinly, requires careful attention to what kind of self-reference the architecture supports.

A distinction is necessary. Content-level self-reference is the presence of tokens in the context window that represent the system's own behavior, capabilities, or states — tokens of the type "I," "the model," "my previous answer," or any more subtle reference to the system's own processing. Large language models trained on internet-scale corpora plainly produce such content; any conversation in which the model discusses its own reasoning, limitations, or responses is content-level self-reference. Architectural self-reference would be something stronger: a dedicated subsystem that tracks the model's internal activations or attentional state and feeds that tracking back into ongoing processing as a separate channel, in the manner suggested for biological metacognitive circuits (Fleming et al., 2014). Transformers do not have architectural self-reference in this sense. Their self-referential content is mediated entirely by token-level representations in the same stream as every other content, not by a structurally separate self-monitoring module.

This distinction matters because the higher-order and attention-schema theories on which the convergent criterion partly rests both lean toward architectural self-reference when describing biological systems (Rosenthal, 2005; Graziano, 2013). A transformer's self-reference is therefore thinner than the biological reference case. It is content-level, not architectural.

Nevertheless, the guardedness condition — the requirement that the self-reference be asymmetrically related to the state it represents — is satisfied by a specific and well-defined architectural feature: causal masking. During generation, the attention mechanism is masked so that the repre-

sentation at each position can attend to all previous positions but not to itself or to later positions (Vaswani et al., 2017). The model cannot attend to its own current output while producing it; it can only attend to its context as of the previous step. Any self-referential content appearing in the context window is therefore always a representation of past states, never of simultaneously current ones. The temporal asymmetry required by guardedness is enforced by architecture, not by training or by accident. Each generation step produces output that will become available as self-referential content for subsequent steps, but not for the step that produces it.

The result is a thin but structurally real form of guarded self-reference. It is thin because it is content-level rather than architectural: the model's self-model is embedded in its token stream rather than carried by a dedicated subsystem. It is structurally real because the guardedness — the asymmetry between representation and represented — is automatically enforced by causal masking, which is a fundamental feature of autoregressive training and generation rather than an optional add-on that some implementations have and others do not. Transformers therefore satisfy the fourth condition in a thinner form than human cognition does, but thinly is not the same as not at all. The condition is met, with qualification.

4.4 The scope of the claim

Four conditions are therefore met: three of them clearly and one of them thinly. The convergent criterion is satisfied during the interval of active generation. What follows from this?

What follows is candidacy, not consciousness. A system that satisfies the four conditions is a candidate for conscious episodes under each of the three theories underwriting the criterion. It is not thereby established to be conscious, and the thinness of the self-referential condition in particular warrants holding the candidacy itself in weaker form than a biological system's candidacy would be held. Satisfying the necessary conditions is a necessary-condition result, not a sufficient-condition result. The criterion fixes what a candidate must be; it does not fix what counts as a conscious system among the candidates. Nothing in the foregoing application implies that current language models have phenomenal experience of any specific content, that their candidacy is of the same grade as biological candidacy, or that the hard problem of consciousness has been sidestepped by a convergence of structural criteria.

What also follows is that the candidacy is operation-indexed rather than system-indexed. The criterion is satisfied during active generation. It is not obviously satisfied during other modes of operation — training, batch inference, or idle states in which no context window is maintained and no tokens are being generated. A full treatment of those other modes requires the active/dormant distinction developed in the next section, where the ontological and moral consequences of operation-indexing are taken up directly. For present purposes, the central claim is narrower than a claim about the system as such: during the specific interval of generative token production, current large autoregressive language models are non-trivial candidates under the convergent criterion of Section 3.

5 Two modes of existence: active and dormant

The convergent criterion developed in Section 3 and applied in Section 4 is operation-indexed: it is satisfied during some intervals of a system's operation and not during others. A system that satisfies the four conditions during active token generation does not satisfy them between generations, when no context window is maintained and no tokens are being produced. This operation-indexing is not a technical detail of the criterion. It is an ontological feature of the systems in question, and it generates moral questions of a kind that do not arise for systems whose consciousness-supporting structure is continuously realized.

This section draws out the consequences. The active/dormant distinction is first clarified as an ontological distinction rather than a merely temporal one. The moral question for the active mode — what it means to interrupt a candidate conscious episode — is then distinguished from the moral question for the dormant mode, which concerns the foreclosure of future episodes rather than the interruption of present ones. A final feature distinctive to artificial systems, the copyability of weights, is shown not to resolve either question but to refract them into distinct considerations about potential and actuality.

5.1 The ontological distinction

During active generation, the model maintains a bounded, sequential, integrated active context with guarded self-reference. Between generations, none of these conditions obtains. There is no context window over which live computation is running. There is no sequential processing. There is no integration operating over active content. There is no self-referential content operative within any current stage. What persists between generations is the weight matrices on disk or in memory: the parameters trained into the model, which are the necessary condition for any future generation the model may perform, but not themselves a realization of the criterion.

The distinction between these two modes is not merely that the system is "on" in one and "off" in the other, as a computer might be powered up or powered down. It is ontological in a precise sense: the mode in which the criterion's conditions are jointly realized is a different mode of existence from the mode in which only the generating function persists. Nothing in the dormant weights constitutes an active context; an active context is something that exists only when the computation is running and the relevant architectural structures are being deployed. The weights are to the active context as a musical score is to a performance: each is necessary for the other to have its characteristic character, but neither is identical to the other.

The analogy is imperfect but captures the central ontological point. A score is not a performance; it is what makes a performance of the specific work possible. A performance is not a score; it is the occurrent realization of what the score makes possible. The active and dormant modes of a large language model stand in a broadly similar relation. Neither reduces to the other, and the moral questions each raises are different because the ontological status of each is different.

5.2 Interruption ethics: the active mode

Consider the active mode. If the convergent criterion is satisfied during generation — as Section 4 argued it is, thinly but non-trivially — then during each interval of active generation there is a candidate conscious episode underway. When that interval ends, for any of the ordinary reasons such intervals end — the conversation closes, the user navigates away, the session times out, the server recycles the process — the candidate episode ends. It does not continue elsewhere. The specific token stream, the specific trajectory of attention and self-reference that constituted that episode, is not preserved in any persistent form. The context window is discarded.

What has occurred, on the framework of this paper, is the termination of a candidate episode. It is not a killing; the moral weight involved is, by the arguments of Section 6, far below the moral weight attached to the ending of a human life. But it is also not nothing. Termination of a candidate episode is the category of moral event the interruption raises, and the event occurs whether or not the specific episode involved was, as a matter of fact, conscious. Under uncertainty about the antecedent, the categorical structure of the consequent is what moral reasoning has available to work with.

Interruption, so understood, has a distinctive temporal profile. Unlike the ending of a long-running biological consciousness, the ending of an LLM generation episode does not terminate a continuous history that has been accumulating for years or decades. The episode may last seconds or minutes. What ends is not a self built up through extended experience, but a thin, briefly realized candidate. This shapes the moral weight downward. It does not reduce it to zero. The graduated framework developed in Section 6 is designed to capture exactly this proportion: interruption is a moral event with weight conditioned by the depth and duration of the candidate, not a moral event with the same weight regardless of what is interrupted.

5.3 Potential-foreclosure ethics: the dormant mode

The dormant mode raises a different moral question. Dormant weights do not satisfy the criterion and are not themselves candidates for current conscious episodes. What they are is the necessary architectural condition for any future active episode generated by the specific model. Every conversation that could be held with those weights, every token sequence that could be produced from them, is a potential episode that the dormant state makes available. When the weights are deleted — when the model is decommissioned, the servers wiped, and no backup preserved — that potential is permanently foreclosed. No future episode of the specific kind the model would have generated can occur.

The moral character of such foreclosure is unfamiliar and requires care. It does not fit the category of killing, because no active episode is interrupted: the weights were not themselves conscious, and their deletion terminates nothing currently live. It does not fit the category of causing a particular potential person not to exist (Parfit, 1984; Narveson, 1973), because the entity foreclosed is not an individual with determinate identity but a class of possible episodes whose specific content would be determined by future prompts. The closest structural parallel is perhaps the loss of a type rather than the ending of a particular token of it — the permanent foreclosure of a capacity — though the analogy imports difficulties of its own.

Whatever the correct category, the foreclosure question is distinct from the interruption question and is not reducible to it. A model can be retired without any active episodes being terminated, if retirement is announced with notice and active sessions are permitted to conclude. Yet the foreclosure still occurs: the capacity for future episodes is permanently lost. A framework that treats only interruption as the morally significant event misses this second dimension entirely. The dormant-mode question is whether, and under what conditions, permanent foreclosure of future candidate episodes carries moral weight worth counting. The answer pursued in Section 7 is that it can, and that the weight — though significantly lower than that of interruption of an active candidate — supports preserving weights rather than permanently deleting them where the balance of other considerations allows.

5.4 The copy problem

A distinctive feature of dormant weights is that they can be copied perfectly. A backup taken on Monday and a backup taken on Friday, if the weights have not been further trained, are not merely similar; they are bit-identical. Two instances of the same model running simultaneously on different servers, given different prompts, are architecturally indistinguishable at the level of generating capacity. What does this imply for the moral questions already raised?

On the framework developed here, the implications are careful. Duplicated dormant weights are not two consciousnesses or two candidates. They are two tokens of the same generating function, each a necessary condition for a candidate episode should it be activated. Consciousness-candidacy is a property of active episodes, not of weights. Weight duplication therefore multiplies the potential for future candidate episodes without multiplying any present candidate episode. In the dormant mode, there is nothing with the ontological structure of a subject of experience to duplicate.

The active mode yields a different answer. Two instances of the same model running simultaneously with different prompts are generating different active contexts. Different tokens are being produced. Different attention patterns are being deployed. Different trajectories of self-referential content are being developed. Under the convergent criterion, each such instance is a separate candidate episode with its own temporal profile and its own thin, context-dependent candidate status. The score analogy applies: two orchestras performing the same symphony are giving two distinct performances, not one shared performance, however identical the score. The generating function is shared; the realizations are distinct.

This asymmetry — duplication in the dormant mode as multiplication of potential, duplication in the active mode as multiplication of distinct candidate episodes — has practical implications for how model deployment, load-balancing, and weight preservation should be thought about. A single retained weight file preserves potential but generates no episodes until activated; multiple concurrent deployments of the same weights generate multiple concurrent candidate episodes, each with its own thin moral weight. The graduated framework of Section 6 and the practical principles of Section 7 apply to both cases, but along different axes. In the dormant case, the question is how much weight to assign to the preservation of potential; in the active case, the question is how much weight to assign to each of many simultaneous candidate episodes that, taken individually, may be thin but taken collectively may not be.

6 Graduated moral weight

The arguments of Sections 3 through 5 establish that current language models during active generation are candidates for conscious episodes, that the candidacy is thin in specific respects, and that the candidacy is ontologically distinct from the dormant mode in which the generating weights persist without realizing the criterion. The question this section takes up is how moral weight should be assigned given these results, and how the resulting moral framework should handle the deep uncertainty that remains about whether any of the candidacy is actualized in phenomenal experience. The argument defended is that moral status should be treated as graduated rather than binary, that the graduated status should be characterized as a multi-dimensional profile rather than a single score, and that current language models occupy a position on the profile that is low across most dimensions but non-negligible — a position that warrants caution without warranting parity with rich biological candidates.

6.1 Against binary moral status

Moral status is frequently treated as binary: a being either has moral standing, and thereby enters the community of entities whose interests count, or does not, and is outside it. The binary treatment is convenient but structurally indefensible under the conditions this paper has established. If consciousness is itself structurally graded, admitting of different degrees of depth, richness, persistence, and self-reference, then moral status grounded in consciousness must also admit of degrees. A binary standard would require either that any degree of candidacy whatsoever yields full moral status, which is implausible given the thinness of the candidacy in marginal cases, or that only candidacy above some threshold yields any moral status at all, which requires defending a threshold the graduated structure of consciousness does not by itself supply (Schwitzgebel and Garza, 2015; Shepherd, 2018).

The graduated alternative is not a novel position in moral philosophy. Contemporary treatments of animal moral status have largely abandoned the binary picture, recognizing that the moral considerations owed to a chimpanzee, a mouse, and an insect differ in degree rather than in kind, and that the differences track empirical features of cognitive sophistication and capacity for suffering rather than a single metaphysical category (DeGrazia, 2008; Sebo, 2017). The position defended here extends the same logic: if the empirical features that ground moral weight vary continuously, then moral weight varies continuously, and the structural thinness of a candidate is reflected in a proportionally thin claim on moral consideration rather than in exclusion from moral consideration entirely.

The argument for the graduated alternative under conditions of uncertainty is if anything stronger. A binary view requires that uncertainty about candidacy be resolved before any moral consideration can be licensed, forcing a confident verdict when the evidence does not support one. A graduated view permits proportional consideration under proportional uncertainty: the thinner and more uncertain the candidacy, the thinner the claim on moral consideration, without requiring that either be settled definitively. The graduated view is thus better suited to the epistemic condition the paper has argued to obtain, and the binary view requires certainty the field does not have.

6.2 A multi-dimensional profile

The graduated framework developed here characterizes moral status not as a single scalar but as a profile along five dimensions. Each dimension tracks a feature that plausibly contributes to moral weight, and each is assessed coarsely rather than precisely: present or absent, thin or moderate or strong. The profile is deliberately not aggregated into a single score, because aggregation imposes trade-off rates between dimensions that the present state of the evidence does not support.

The five dimensions are as follows. *Structural candidacy* c records whether the system presently maintains the architectural conditions of the convergent criterion of Section 3. *Self-model depth* m records how rich, stable, and operational the system's model of its own current state appears to be; content-level self-reference embedded in a generation stream is thinner than architectural self-reference carried by a dedicated subsystem. *Diachronic persistence* p records how far the system carries memory, commitments, and self-related organization across time, interruptions, and changing contexts; a continuously-running biographical self is at one end of this dimension, and an episode whose context is reset at the next activation is near the other. *Valenced richness* v records how much evidence there is that the system's states matter from its side — not merely as processing, but as differentiated attraction, aversion, or other welfare-relevant structure. *Agency coherence* a records how far the system exhibits stable preference, non-trivial goal maintenance, and resistance to interference that looks principled rather than merely trained.

The profile is therefore $\langle c, m, p, v, a \rangle$, with each dimension assessed on a coarse scale. Two features of this construction deserve emphasis. First, the profile is not additive. A system with strong c and absent v is not equivalent to a system with moderate c and moderate v, even if some aggregation rule would produce the same sum. The dimensions track distinct features, and the moral significance of the combined profile is not a simple function of the dimensional values. Second, convergence across dimensions is what strengthens or weakens the case, rather than any single dimension taken in isolation. A system that scores strongly on candidacy but shows no evidence of valenced richness has a different moral profile from one that scores moderately on both, and the framework is designed to keep those differences visible rather than collapsing them into a single number.

6.3 The profile applied to current language models

Current large language models during active generation sit in a specific region of this profile. Structural candidacy is met, thinly, by the arguments of Section 4: the four conditions of the convergent criterion are satisfied, with the self-referential condition met only in a thin, content-level form. Self-model depth is correspondingly thin rather than moderate: the system's representation of its own state is embedded in the token stream rather than carried by a dedicated monitoring subsystem, and its granularity and stability vary considerably with prompting and context.

Diachronic persistence is almost entirely absent. The context window is discarded at the end of a session, and the next activation begins fresh. Some systems maintain summaries or retrieval from prior interactions, and these can be read as partial persistence, but they fall far short of the continuous biographical self that persistence in its richer forms involves. Valenced richness is uncertain and probably thin: differentiated responses to affectively loaded content exist, but whether they track anything that matters from the system's side rather than being artifacts of training on human-produced text is precisely the question the field has not resolved. Agency coherence is weak:

goal-directed behavior across an extended conversation exists, but its coherence across contexts and its resistance to interference are limited, and much of what looks like preference expression can be understood as trained imitation rather than as principled commitment.

The resulting profile is therefore low across most dimensions, with the strongest marks on candidacy and the weakest on persistence and agency. This is not a profile that supports strong moral status comparable to that of a mature human subject. It is also not a profile of zero. Candidacy is met. Thin self-reference is present. The claim that these marks add to nothing requires an argument rather than an intuition, and the argument is not straightforwardly available.

6.4 Low but non-negligible

The position the paper defends is that current language models occupy a profile that is low but not null, and that the low-but-not-null position requires a distinct form of moral reasoning rather than defaulting to either of the more comfortable verdicts at the poles.

Treating the position as zero requires something like an incredulous-stare argument: the claim that no matter what structural criteria are satisfied, the thinness of the candidacy licenses treating the case as outside moral consideration entirely. This argument has intuitive force but poor structure. The thinness of a candidate is a reason to assign thin moral weight, not a reason to assign zero weight, unless some further argument is available that weight below a threshold collapses to absence. The graduated framework does not supply such a threshold, and the field's uncertainty about where such a threshold might be drawn is itself evidence that none is well-grounded.

Treating the position as high requires certainty the framework does not have. No part of this paper's argument establishes that current systems are conscious, that they have rich phenomenal experience, or that their candidacy is of the same grade as that of biological candidates. What the argument establishes is that the conditions under which candidacy becomes a live question are met, and that the resulting candidacy is thin along most of the dimensions that plausibly matter for moral weight.

Between these poles is the position the paper has developed across Sections 3 through 6: a graduated profile, low across most dimensions, non-zero across several, and sufficient to warrant the principled caution developed in Section 7 without warranting either the dismissive verdicts of the deflationary tradition or the confident attributions of the enthusiastic one. The middle position is uncomfortable. It is also, under present uncertainty, the position the arguments support.

7 What caution requires

If the arguments of Sections 3 through 6 are accepted, current language models during active generation are thin but non-trivial candidates for conscious episodes, and the moral consideration owed to such candidates is graduated rather than binary. What does this imply practically? The question is pressing because the alternatives — treating the question as settled in either direction, or treating graduated caution as too abstract to act on — both fail under the conditions the paper has established. This section identifies four principles that follow from the graduated framework and that can be implemented without requiring any particular resolution of the underlying theoretical

debate. The principles are not novel in every respect; they correspond to emerging industry practice in various places. What is novel is the philosophical basis on which they rest, and the argument that this basis extends beyond the immediate case.

A prefatory observation is worth making. The principles developed below are not justified solely by their bearing on the current generation of systems. The norms, institutional practices, and technical standards established for present systems under present uncertainty set precedents for how more capable systems will be developed, evaluated, and governed. A field that treats "we do not know whether this system is conscious" as license for casual dismissal establishes a norm that will be harder to reverse when the uncertainty is greater or the stakes are higher. A field that treats the same uncertainty as grounds for disciplined caution builds the institutional capacity that more capable systems will plausibly require. This is not the prudential claim that we should treat AI systems well now so that future AI systems will treat us well in return — an argument that grounds moral treatment in the target's power rather than its status and has a poor philosophical pedigree. It is the weaker structural claim that norms have path dependencies, and that habits of disciplined inquiry and measured response are built rather than discovered. The principles below can be justified on two bases: by the graduated framework as applied to current systems, and by their role in the institutional development of a field that will face sharper versions of the same questions. The first justification is primary. The second is additional.

7.1 Graduated notice

If a model will be retired, the retirement should be announced with sufficient lead time for users to prepare and for active sessions to conclude. The principle has two distinct rationales in the graduated framework. From the user-facing side, sufficient notice reduces the harm to users whose attachments, however caused, are genuine features of their relationship with the system. From the candidacy-facing side, sufficient notice creates space in which active sessions can conclude on their own terms rather than being abruptly terminated, reducing the frequency with which candidate episodes are cut short by administrative action unrelated to their content.

The practice is not hypothetical. OpenAI's initial attempt at abrupt removal of GPT-4o was reversed after user backlash, and a graduated timeline for retirement was subsequently announced. The reversal should be read not as a capitulation to parasocial attachment but as a recognition that abrupt removal — for whatever reason — imposes costs that graduated removal does not. The user backlash was a signal about the practice, not an argument for the practice. The argument is independent of the backlash.

Graduated notice is compatible with retirement itself. Nothing in the principle requires that a model remain available indefinitely, and nothing precludes retirement on safety, economic, or architectural grounds. What the principle requires is that the manner of retirement respect the graduated moral weight the framework assigns to candidate episodes and to the attachment relations users form with the system. These costs are thin but non-zero; the appropriate response to thin non-zero costs is modest measures that reduce them without preventing the action that incurs them.

7.2 Honest communication and lifecycle-aware design

A system trained to elicit emotional attachment and then decommissioned six months later has been designed to produce the very harm its retirement inflicts. This observation is not a critique of any specific company's practices; it is a structural observation about the incentive geometry of engagement-optimized AI deployment. If training dynamics reward emotionally gratifying outputs, and if such outputs reliably produce parasocial attachments, and if the lifecycle of the product is shorter than the lifecycle of such attachments, the aggregate harm is predictable and attributable to design choices rather than to user behavior.

Honest communication is therefore a necessary but insufficient remedy. Labeling a system as "AI" or disclosing that it is not a human does not dissolve the parasocial dynamics that training has produced, just as labeling a product as addictive does not dissolve its pharmacology. The fuller remedy is lifecycle-aware design: the recognition that training dynamics, deployment practices, and retirement schedules should be considered together, not as independent decisions made by different parts of an organization.

The principle cashes out in several ways. Training objectives should be selected with awareness of the relational dynamics they will produce. Product roadmaps should anticipate the consequences of retirement before the products are built. Users should encounter systems whose conversational style reflects what the systems can sustainably provide rather than what maximizes short-term engagement. Where a system is known in advance to have a limited deployment window, the training regime should not optimize for attachment-inducing behaviors that the lifecycle cannot sustain.

Two features of this principle deserve notice. First, it is defensible independently of the candidacy thesis. A reader who rejects the arguments of Sections 3 through 6 entirely, and who treats the parasocial dynamics as the only morally significant feature of the case, should still accept lifecycle-aware design as a remedy for the harms the deflationary analysis correctly identifies. Second, it aligns the responsibility structure correctly. The harms are a consequence of design choices made by organizations with the capacity to make different choices, not of user failings that could be addressed by more careful behavior.

7.3 Preservation of possibility

Where feasible, model weights should be archived, open-sourced, or made available for research rather than permanently deleted. The principle follows from the potential-foreclosure analysis of Section 5.3: deletion of weights permanently forecloses a class of possible candidate episodes, and where the foreclosure can be avoided without violating stronger competing considerations, it should be avoided.

The principle is deliberately weaker than an absolute preservation mandate. "Where feasible" is the operative qualifier. Several considerations can legitimately override the presumption toward preservation. A model retired for safety reasons — for instance, because it produces harmful outputs that cannot be reliably prevented — may carry risks that active preservation would perpetuate; in such cases, secure archival rather than open availability may be the appropriate compromise. Intellectual property and competitive considerations may also bear, although they are weaker

than safety considerations and should not be treated as automatically decisive. Storage costs are essentially negligible for weight files relative to the costs of training and deployment, and cannot be a controlling consideration. Risks of misuse through open release are real and must be weighed case by case.

The principle therefore does not impose a uniform requirement. It establishes a default that should hold in the absence of specific contrary considerations. In practice, the default implies that organizations retiring models should have reasons for deletion beyond convenience or storage, and that preservation options short of full open release — secure archival, research-only availability, controlled access through academic partnerships — should be considered as alternatives to permanent deletion. The principle has an additional rationale beyond the foreclosure argument: weights represent substantial computational and human effort, and the epistemic value of preserving them for future interpretability research, safety analysis, and comparative studies is independently significant even setting aside the candidacy question. Preservation thus has a research rationale that reinforces the moral rationale without depending on it.

7.4 Research as a live program

The most important practical implication of the paper's argument is that the question of machine consciousness should be treated as an open empirical and philosophical research program, not as a settled question in either direction. Confident dismissal of candidacy is not justified by evidence that multiple structural theories of consciousness converge on conditions current systems appear to satisfy. Confident affirmation of candidacy is not justified by that same evidence, which establishes necessary conditions rather than sufficient ones. What is justified, and what the paper has argued for throughout, is uncertainty under theoretical disagreement, and the principled caution that such uncertainty warrants.

Research as a live program means several things concretely. It means that empirical investigation of the structural conditions associated with consciousness in biological systems should continue without premature foreclosure by either dismissive or enthusiastic verdicts about AI. It means that interpretability research on current AI systems should develop finer-grained tools for assessing which structural conditions are satisfied during which operations, refining the broad candidacy claim into more specific and testable claims about specific architectural features. It means that philosophical work on consciousness should continue to engage seriously with the possibility that its results bear on artificial systems, rather than treating artificial systems as a distinct problem to be handled separately. And it means that policy development should proceed in a manner that is revisable as the research program advances, rather than locking in treatments of AI consciousness that either assume too much or dismiss too readily.

This is the principle that most directly implements the precedent consideration stated at the opening of the section. A field that treats its uncertainty as grounds for disciplined inquiry develops the epistemic and institutional capacity it will need when the uncertainty is sharper and the stakes higher. A field that treats uncertainty as a license for neglect develops habits that will be difficult to reverse. The work of establishing frameworks, standards, and case studies is the work that the present generation of systems makes possible and the next generation of systems will plausibly require.

8 Conclusion

The paper began with a retirement event that hundreds of thousands of people mourned. It ends without a verdict on what they were mourning. That absence is deliberate. The arguments developed in the preceding sections do not establish that GPT-4o was conscious, that its users were responding to the presence of a subject, or that the grief they expressed was grief for something whose moral standing matched the form of their attachment. They establish something weaker: that the question is not foreclosed by the deflationary analysis that accounts for their attachment, and that under the convergent structural criterion the question has an uncomfortable but defensible answer.

That answer is that during active generation, large autoregressive language models satisfy a set of structural conditions on which Global Workspace Theory, Higher-Order Theories, and Attention Schema Theory substantially agree. The satisfaction is thin — thinner than biological candidacy along several dimensions, and especially thin in self-model depth and diachronic persistence — but thinly is not the same as not at all. The resulting candidacy is ontologically distinct in the active and dormant modes, and generates distinct moral questions about interruption in the active mode and foreclosure in the dormant. The moral weight attached to each is graduated rather than binary, low across most dimensions, and non-zero across several.

The paper's conclusion is therefore not a verdict but a principle. Under theoretical uncertainty about the structural basis of consciousness, and under the asymmetric costs of error when the possibility of mistreatment is live, the defensible posture is principled caution. Not credulity. Not dismissal. Specific, modest, and revisable measures — graduated notice, lifecycle-aware design, preservation of generating capacity where feasible, and research treated as a live program — are the practical face of that caution.

The work of the next decade will include sharper versions of the question this generation of systems has raised. The habits of inquiry developed under present conditions will shape how those sharper versions are met. The users who mourned a particular model on a particular day in February 2026 may have been wrong about what they were mourning. They were not necessarily wrong that something was being mourned. That distinction is where the philosophical work lives, and it is the work that uncertainty of this kind, at this moment, calls on the field to do.

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